

ID:- 2024 DC 04030

NAME:- AKASH MALIK

Ques:- Perform Principle Component Analysis (PCA) on multivariate data to reduce dimensionality and analyse structure in health-related features.

Person	Age	BMI	Blood Pressure	cholesterol Daily walk steps	steps hours of sleeps	<u>sleeps</u>
P1	25	22	118	180	8000	7.0
P2	32	27	130	210	6000	6.5
P3	45	30	145	220	5000	6.0
P4	29	24	125	190	7500	7.5
P5	35	29	140	200	6200	6.8
P6	40	31	150	230	4800	6.2
P7	27	23	120	175	8200	7.2
P8	50	32	160	240	4500	5.8
P9	38	28	135	195	6700	6.6
P10	31	26	128	185	7100	6.9

(A) Pre processing

(1) For Age

$$\text{Mean } (\mu) \Rightarrow \frac{25 + 32 + 45 + 29 + 35 + 40 + 27 + 50 + 38 + 31}{10} = 35.2000$$

$$\text{Standard deviation } (\sigma) \Rightarrow \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \mu)^2}$$

$$\Rightarrow \sqrt{\frac{1}{9} \times \left\{ (25-35.2)^2 + (32-35.2)^2 + (45-35.2)^2 + (29-35.2)^2 + (35-35.2)^2 + (40-35.2)^2 + (27-35.2)^2 + (50-35.2)^2 + (38-35.2)^2 + (31-35.2)^2 \right\}}$$

$$\Rightarrow \sqrt{64.8444} \Rightarrow 8.0526$$

Similarly we'll calculate for [Similar logic is used below & direct value is written]
BMI

$$\text{Mean } (\mu) \Rightarrow 27.2000$$

$$\text{Standard deviation } (\sigma) \Rightarrow 3.4254$$

Similarly, Blood Pressure

$$\text{Mean } (\mu) \Rightarrow 135.1000$$

$$\text{Standard deviation } (\sigma) \Rightarrow 13.5929$$

Similarly cholesterol,

$$\text{Mean } (\mu) \Rightarrow 202.5000$$

$$\text{Standard deviation } (\sigma) \Rightarrow 21.8899$$

Similarly Daily walk steps

$$\text{Mean } (\mu) \Rightarrow 6400.0000$$

$$\text{Standard deviation } (\sigma) \Rightarrow 1329.9958$$

Similarly Hours of sleep

$$\text{Mean } (\mu) \Rightarrow 6.6500$$

$$\text{Standard deviation } (\sigma) \Rightarrow 0.5380$$

(A)(2)

Calculations for Person (P1) $\rightarrow$

$$\text{Age:- } (25 - 35.2) / 8.0526 \Rightarrow -1.2667$$

$$\text{BMI} \Rightarrow (22 - 27.2) / 3.4254 \Rightarrow -1.5181$$

$$\text{Blood pressure} \Rightarrow (118 - 135.1) / 13.5929 \Rightarrow -1.2580$$

$$\text{cholesterol} \Rightarrow (180 - 202.5) / 21.8899 \Rightarrow -1.0279$$

$$\text{Daily walk steps} \Rightarrow (8000 - 6400) / 1329.9958 \Rightarrow 1.2030$$

$$\text{Hours of sleep} \Rightarrow (7.0 - 6.65) / 0.538 \Rightarrow 0.6506$$

Calculation for Person (P2) :-

(3)

Age $\Rightarrow (32 - 35.2) / 8.0526 \Rightarrow -0.397$
BMI $\Rightarrow (27 - 27.2) / 3.4254 \Rightarrow -0.058$
Blood Pressure $\Rightarrow (130 - 135.1) / 13.5929 \Rightarrow -0.375$
cholesterol $\Rightarrow (210 - 202.5) / 21.8899 \Rightarrow 0.343$
steps $\Rightarrow (6000 - 6400) / 1329.9958 \Rightarrow -0.301$
sleep $\Rightarrow ~~6.5~~ (6.5 - 6.65) / 0.538 \Rightarrow -0.279$

P₃

Age $\Rightarrow (45 - 35.2) / 8.0526 \Rightarrow 1.217$
BMI $\Rightarrow (30 - 27.2) / 3.4254 \Rightarrow 0.817$
Blood Pressure $\Rightarrow (145 - 135.1) / 13.5929 \Rightarrow 0.728$
cholesterol $\Rightarrow (220 - 202.5) / 21.8899 \Rightarrow 0.799$
steps $\Rightarrow (5000 - 6400) / 1329.9958 \Rightarrow -1.053$
sleep $\Rightarrow (6.0 - 6.65) / 0.538 \Rightarrow -1.208$

P₄

Age $\Rightarrow (29 - 35.2) / 8.0526 \Rightarrow -0.770$
BMI $\Rightarrow (24 - 27.2) / 3.4254 \Rightarrow -0.934$
BP $\Rightarrow (125 - 135.1) / 13.5929 \Rightarrow -0.743$
cholesterol $\Rightarrow (190 - 202.5) / 21.8899 \Rightarrow -0.571$
steps $\Rightarrow (7500 - 6400) / 1329.9958 \Rightarrow 0.827$
sleep $\Rightarrow (7.5 - 6.65) / 0.538 \Rightarrow 1.580$

P₅

Age $\Rightarrow (35 - 35.2) / 8.0526 \Rightarrow -0.025$
BMI $\Rightarrow (29 - 27.2) / 3.4254 \Rightarrow 0.525$
BP $\Rightarrow (140 - 135.1) / 13.5929 \Rightarrow 0.360$
cholesterol $\Rightarrow (200 - 202.5) / 21.8899 \Rightarrow -0.114$
steps $\Rightarrow (6200 - 6400) / 1329.9958 \Rightarrow -0.150$
sleep $\Rightarrow (6.8 - 6.65) / 0.538 \Rightarrow 0.279$

P₆

$$\text{Age} \Rightarrow (40 - 35.2) / 8.0526 \Rightarrow 0.596$$

$$\text{BMI} \Rightarrow (31 - 27.2) / 3.4254 \Rightarrow 1.109$$

$$\text{BP} \Rightarrow (150 - 135.1) / 13.5929 \Rightarrow 1.096$$

$$\text{cholesterol} \Rightarrow (230 - 202.5) / 21.8899 \Rightarrow 1.256$$

$$\text{steps} \Rightarrow (4800 - 6400) / 1329.9958 \Rightarrow -1.203$$

$$\text{sleep} \Rightarrow (6.2 - 6.65) / 0.5380 \Rightarrow -0.836$$

P₇

$$\text{Age} \Rightarrow (27 - 35.2) / 8.0526 \Rightarrow -1.018$$

$$\text{BMI} \Rightarrow (23 - 27.2) / 3.4254 \Rightarrow -1.226$$

$$\text{BP} \Rightarrow (120 - 135.1) / 13.5929 \Rightarrow -1.111$$

$$\text{cholesterol} \Rightarrow (175 - 202.5) / 21.8899 \Rightarrow -1.256$$

$$\text{steps} \Rightarrow (8200 - 6400) / 1329.9958 \Rightarrow 1.353$$

$$\text{sleep} \Rightarrow (7.2 - 6.65) / 0.5380 \Rightarrow 1.022$$

P₈

$$\text{Age} \Rightarrow (50 - 35.2) / 8.0526 \Rightarrow 1.838$$

$$\text{BMI} \Rightarrow (32 - 27.2) / 3.4254 \Rightarrow 1.401$$

$$\text{BP} \Rightarrow (160 - 135.1) / 13.5929 \Rightarrow 1.832$$

$$\text{cholesterol} \Rightarrow (240 - 202.5) / 21.8899 \Rightarrow 1.713$$

$$\text{steps} \Rightarrow (4500 - 6400) / 1329.9958 \Rightarrow -1.429$$

$$\text{sleep} \Rightarrow (5.8 - 6.65) / 0.5380 \Rightarrow -1.580$$

P₉

$$\text{Age} \Rightarrow (38 - 35.2) / 8.0526 \Rightarrow 0.348$$

$$\text{BMI} \Rightarrow (28 - 27.2) / 3.4254 \Rightarrow 0.234$$

$$\text{BP} \Rightarrow (135 - 135.1) / 13.5929 \Rightarrow -0.007$$

$$\text{cholesterol} \Rightarrow (195 - 202.5) / 21.8899 \Rightarrow -0.343$$

$$\text{steps} \Rightarrow (6700 - 6400) / 1329.9958 \Rightarrow 0.226$$

$$\text{sleep} \Rightarrow (6.6 - 6.65) / 0.5380 \Rightarrow -0.093$$

P₁₀

$$\text{Age} \Rightarrow (31 - 35.2) / 8.0526 \Rightarrow -0.522$$

$$\text{BMI} \Rightarrow (26 - 27.2) / 3.4254 \Rightarrow -0.350$$

$$\text{BP} \Rightarrow (128 - 135.1) / 13.5929 \Rightarrow -0.522$$

$$\text{cholesterol} \Rightarrow (185 - 202.5) / 21.8899 \Rightarrow -0.799$$

$$\text{steps} \Rightarrow (7100 - 6400) / 1329.9958 \Rightarrow 0.526$$

$$\text{sleep} \Rightarrow (6.9 - 6.65) / 0.5380 \Rightarrow 0.465$$

Standardized data matrix :-

⑤

Person	Age	BMI	BP	cholesterol	steps	sleep
P ₁	-1.267	-1.518	-1.258	-1.028	1.203	0.651
P ₂	-0.397	-0.058	-0.375	0.343	-0.301	-0.279
P ₃	1.217	0.817	0.728	0.799	-1.053	-1.208
P ₄	-0.770	-0.934	-0.743	-0.571	0.827	1.580
P ₅	-0.025	0.525	0.360	-0.114	-0.150	0.279
P ₆	0.596	1.109	1.096	1.256	-1.203	-0.836
P ₇	-1.018	-1.226	-1.111	-1.256	1.353	1.022
P ₈	1.838	1.401	1.832	1.713	-1.429	-1.580
P ₉	0.348	0.234	-0.007	-0.343	0.226	-0.093
P ₁₀	-0.522	-0.350	-0.522	-0.799	0.526	0.465

Now, for the covariance matrix

$$\text{Covariance formula } (C_{ij}) \Rightarrow \frac{1}{(n-1)} \times \sum (Z_{Ki} \times Z_{Kj})$$

$$(n-1) = 9$$

Z = standardized data matrix

	Age	BMI	BP	CHOLE.	STEPS	SLEEP
Age	1	0.933	0.760	0.679	-0.515	-0.616
BMI	0.933	1	0.838	0.766	-0.514	-0.618
BP	0.760	0.838	1	0.939	-0.382	-0.442
CHOLE.	0.679	0.766	0.939	1	-0.422	-0.448
STEPS	-0.515	-0.514	-0.382	-0.422	1	0.919
SLEEP	-0.616	-0.618	-0.442	-0.448	0.919	1

(B) PCA Computation

(3) Eigen values & Eigen vectors

To find the eigen values

Step 1:- Initialization

Let initial vector $v^{(0)} = [1, 1, 1, 1, 1, 1]$

Normalized $[Av^{(0)}]$ {Note:- A = Covariance matrix in part (B)}

after normalized by the ~~max~~ norm

$$Av^{(0)} \Rightarrow \begin{bmatrix} 2.191 \\ 2.405 \\ 2.688 \\ 2.489 \\ 2.086 \\ -0.255 \end{bmatrix}$$

max value = 2.688

... Continued

$$V^{(1)} \Rightarrow [2.91, 2.405, 2.688, 2.489, 0.086, -0.255]$$

$$2.688$$

$$\Rightarrow [0.815, 0.895, 1, 0.926, 0.032, -0.095]$$

Now,

Iteration 2 $\rightarrow$

$$\Rightarrow AV^{(1)}$$

$$\Rightarrow \begin{bmatrix} 3.086 \\ 3.246 \\ 3.246 \\ 3.108 \\ -1.708 \\ -2.019 \end{bmatrix}$$

Similarly max $\Rightarrow 3.246$

$$V^{(2)} \Rightarrow [0.950, 1.000, 1.000, 0.957, -0.526, -0.622]$$

Now, after several iterations,
the vector converges to

$$V_1 = [0.448, 0.472, 0.472, 0.452, -0.249, -0.294]$$

The eigen value (λ_1) $\Rightarrow 3.936$

For Second Eigen value

$$A_2 = A - \lambda_1 V_1 V_1^T$$

$\downarrow$
Covariant matrix

$\downarrow$
deflate the matrix
to exact λ_2, V_2 exactly
by removing values of V_1 .

On solving the calculations we get A_2 .

~~A_2~~

Initialize $V^{(0)} = [1, 1, 1, 1, 1, 1]$

$$A_2 V^{(0)} =$$

$$\begin{bmatrix} 0.209 & 0.101 & -0.072 & -0.118 & -0.075 & -0.146 \\ 0.101 & 0.123 & -0.039 & -0.074 & -0.052 & -0.072 \\ -0.072 & -0.039 & 0.123 & 0.074 & 0.080 & 0.104 \\ 0.118 & -0.074 & 0.074 & 0.195 & 0.020 & 0.075 \\ -0.075 & -0.052 & 0.080 & 0.020 & 0.757 & 0.632 \\ -0.146 & -0.072 & 0.104 & 0.075 & 0.632 & 0.661 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -0.101 \\ -0.013 \\ 0.270 \\ 0.172 \\ 1.362 \\ 1.254 \end{bmatrix}$$

$$V^{(1)} \Rightarrow [-0.074, -0.010, 0.198, 0.126, 1.000, 0.921]$$

Similarly by some iterations

$$V_2 = [0.441, 0.378, 0.243, 0.266, 0.708, 0.605]$$

$$\lambda_2 = 1.497$$

Similarly,

$$A_3 = A_2 - \lambda_2 V_2 V_2^T$$

by doing the same calculations, we get the remaining values.

So, Now, Eigenvalues

$$\lambda_1 \Rightarrow 3.936$$

$$\lambda_2 \Rightarrow 1.497$$

$$\lambda_3 \Rightarrow 0.394$$

$$\lambda_4 \Rightarrow 0.104$$

$$\lambda_5 \Rightarrow 0.045$$

$$\lambda_6 \Rightarrow 0.024$$

Eigen vectors

$$V_1 \Rightarrow [0.448, 0.472, 0.472, 0.452, -0.249, -0.294]$$

$$V_2 \Rightarrow [0.441, 0.378, 0.243, 0.266, 0.708, 0.605]$$

$$V_3 \Rightarrow [0.312, 0.203, -0.292, -0.382, 0.372, 0.406]$$

$$V_4 \Rightarrow [0.224, -0.209, -0.455, 0.372, -0.234, 0.123]$$

$$V_5 \Rightarrow [0.382, -0.569, 0.225, -0.098, 0.229, -0.245]$$

$$V_6 \Rightarrow [0.448, -0.372, 0.219, -0.539, -0.385, 0.298]$$

④

Rank the principle component

PC	EIGEN VALUES	% Variance	Cumulative Percentage variance
PC1	3.936	65.6%	65.60%
PC2	1.497	24.95%	90.55%
PC3	0.394	6.57%	97.12%
PC4	0.104	1.73%	98.85%
PC5	0.045	0.75%	99.60%
PC6	0.024	0.40%	100%

(5) components for 95% variance
 components needed for $\geq 95\%$
 components needed for greater than 95%
 Variance $\Rightarrow \underline{3}$

cumulative variance for 3 variables $\Rightarrow 97.12\%$

(C) (6)
 $\rightarrow$ Projection onto First 3 PCs

$$PC1 \Rightarrow z \cdot v_1 \quad PC2 \Rightarrow z \cdot v_2, \quad PC3 \Rightarrow z \cdot v_3$$

$z \Rightarrow$ standardized data vector
 $v_i \Rightarrow$ Eigen vector

For P1

$$\begin{aligned} PC1 &\Rightarrow (-1.266)(0.448) + (-1.518)(0.472) + \\ &(-1.258)(0.472) + (-1.028)(0.452) + \\ &(1.203)(-0.249) + (0.651)(-0.294) \\ &\Rightarrow -2.832 \end{aligned}$$

$$\begin{aligned} PC2 &\Rightarrow (-1.266)(0.441) + (-1.518)(0.378) + \\ &(-1.258)(0.243) + (-1.028)(0.266) + \\ &(1.203)(0.708) + (0.651)(0.605) \\ &\Rightarrow -0.466 \end{aligned}$$

$$\begin{aligned} PC3 &\Rightarrow (-1.266)(0.312) + (-1.518)(0.203) + \\ &(-1.258)(-0.292) + (-1.028)(-0.382) + \\ &(1.203)(0.372) + (0.651)(0.406) \\ &\Rightarrow 0.769 \end{aligned}$$

Similarly we can calculate the value of P2 to P10.
 by applying the same formula applied above.

Projection dataset table

For P2

$$\begin{aligned} PC1 &\Rightarrow (-0.397)(0.448) + (-0.058)(0.472) + \\ &\quad (-0.375)(0.472) + (0.343)(0.452) + \\ &\quad (-0.301)(-0.249) + (-0.279)(-0.294) \\ &\Rightarrow -0.070 \end{aligned}$$

$$PC2 \Rightarrow -0.579$$

$$PC3 \Rightarrow -0.382$$

similarly calculation for others.

Person	PC1	PC2	PC3
P1	-2.832	-0.466	0.769
P2	-0.070	-0.579	-0.382
P3	2.253	-0.241	-0.855
P4	-2.066	0.515	0.954
P5	0.310	0.307	0.095
P6	2.421	-0.076	-1.176
P7	-2.765	0.060	1.155
P8	3.944	-0.728	-1.505
P9	0.079	0.252	0.336
P10	-1.273	-0.049	0.608

(D) Analysis

(9) 97.12% variance of $m=3$

:- PC1 (Health Risk Factor):-

Positive $\rightarrow$ on Age (0.448), BMI (0.472), BP (0.472) & cholesterol (0.452).

Negative $\rightarrow$ Steps (-0.249) & sleep (-0.294)

Means Higher the value of PC1 mean older age, higher BMI, high blood pressure, high cholesterol, lower physical activity (steps) & less sleep.

:- PC2 (Activity & Reset Factor):-

positive $\Rightarrow$ high Loading = Steps & Sleep

moderate Loading = Age, BMI, BP & cholesterol

Means, Higher the value of PC2 more daily steps, daily more sleep accompanied by moderately high Age, BMI, BP & cholesterol.

:- PC3 (Contrast factor):-

positive:- Age, BMI, steps & sleep

negative:- BP & cholesterol

Means, higher the value of PC3 then older age, higher BMI, more steps & more sleep. But lower blood pressure & cholesterol.

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Yes, correlation between physical activity (walk, sleep)
& health indicators (Age, BMI, BP, cholesterol)

PC1 → Shows a strong negative correlation between
physical activity & health risk indicators.

whereas, PC2 shows a positive correlation between physical
activity and all health indicators, including steps & sleep.

PC3 also shows a positive relationship between physical
activity (steps) & sleep, but with a negative relationship
between BP & cholesterol.